Siqi Yao B.Sc.

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Research Statement

My main research interests lie in AI for Science, especially its applications in materials science and fluid simulation.

Education

Department of Mathematics, Hong Kong University of Science and Technology
Hong Kong SAR, China
Incoming Ph.D. Student in Mathematics 2026.09 –

- Advisor: Prof. Xiang Yang.

School of Data Science, The Chinese University of Hong Kong, Shenzhen Shenzhen, China
B.Sc. in Data Science and Big Data Technology 2022.09 – 2026.07

- CGPA: 3.89/4.00; Major Rank: 7/150.
- Dean's List: AY2022–2023, AY2023–2024, AY2024–2025.
- Academic Performance Scholarship, Class B: AY2023–2024, AY2024–2025.

University of California, Berkeley Berkeley, CA, USA
Berkeley Global Access Program, Visiting Student 2024.08 – 2024.12

- GPA: 4.00.

Experiences

MPM-Based Fluid Simulation Algorithms for Mobile Physics Engines | Supervised by Dr. Zhihao Shen (Huawei) and Prof. Xiang Yang (HKUST) 2026.01 – Present

- Studied fluid simulation algorithms, especially the Material Point Method (MPM).
- Reproduced MLS-MPM using taichi_mpm and developed a dam-break simulation demo to validate the algorithm pipeline.
- Reproduced IQ-MPM with ZIRAN and integrated its C++ simulation backend into a HarmonyOS application through DevEco Studio, with ongoing work on mobile-side runtime adaptation and performance optimization.

Deep Generative Models for Boltzmann Distribution Sampling | Supervised by Prof. Kai Zhou (CUHK-SZ) 2024.08 – 2025.08

- Studied the theoretical foundations of score-based and flow-based generative models, with emphasis on their connections to statistical physics and Boltzmann distribution sampling.
- Analyzed *Energy-Based Diffusion Generator for Efficient Sampling of Boltzmann Distributions* and reproduced its main numerical experiments.
- Explored a stochastic-interpolation-based modification to the original sampling procedure.

Learning-Augmented Wind Power Optimization | Supervised by Dr. Jingyi Zhao (Shenzhen Research Institute of Big Data) 2024.12 – 2025.06

- Studied Learning to Optimize (L2O) methods and mixed-integer nonlinear programming (MINLP) formulations for data-driven decision-making under operational constraints.
- Implemented a *predict-then-optimize* pipeline for a real-world wind power optimization problem in Shanxi Province, integrating MoE-based forecasting with MINLP-based constrained optimization.

Projects

Comparing Traditional Methods and Diffusion Models for Hybrid Image and Visual Anagram Generation Tasks | 2025 Summer

- Implemented hybrid image and visual anagram generation using diffusion model and traditional methods such as image pyramids.
- Systematically compared the performance using multiple metrics such as visual quality and CLIP similarity score.

Building Interpretable Emotional Dialogue Agents via Chain-of-Thought Reasoning | 2025 Spring

- Developed a unified framework for machine emotional intelligence that integrates emotion recognition, cause inference, shift detection, and dialogue generation using Chain-of-Thought (CoT) reasoning.

ICU Simulation and Resource Optimization | 2024 Fall

- Utilize simulation frameworks to optimize resources in the Intensive Care Unit (ICU), including critical components such as beds and nursing staff.
- Key components include ICU queue simulation, nursing staff workflow simulation and various optimization methods such as exhaustive search, heuristics, and Pareto frontier-based method.

Internships

Shenzhen Guanghuiyuan Asset Management Co., Ltd. | Shenzhen, China 2023.07 - 2023.08

- Studied and participated in value investment research.

Skills

Languages: Mandarin, English.

Programming: `Python`, `TensorFlow`, `PyTorch`, `C`, `C++`, `Linux`.

Others: Solid math foundation. Eager to learn new things.

Last updated: 2026.05.14